

Properties of Materials

Different materials have different **physical** and **mechanical** properties. I kid you not.

Mechanical Properties — Learn 'em All...

You need to be **totally familiar** with these terms — if you start getting strength and hardness mixed up, or get confused between malleability and ductility, you'll **drop marks** all over the place.

STRENGTH:

- 1) A material's strength is a measure of its ability to **withstand forces** without **breaking**.
- 2) **Tensile strength** resists **pulling** forces — e.g. the rope in a tug-of-war.
- 3) **Compressive strength** resists **squashing** forces — e.g. bridge supports.
- 4) **Bending strength** resists forces trying to **bend** — e.g. surfboards.
- 5) **Shear strength** resists strong **sliding forces** — e.g. a rivet needs to withstand shear forces.
- 6) **Torsional strength** resists **twisting forces** — e.g. drill bits need to withstand twisting forces.



HARDNESS:

- 1) The ability to withstand **abrasive** wear and tear, **denting** and **bending**.
- 2) Very important for tools that cut, like **files** and **drills**.

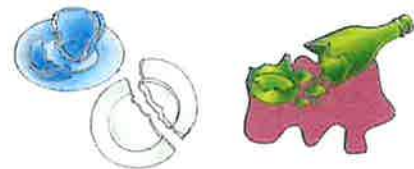


PLASTICITY:

- 1) If a product can change shape **permanently, without breaking or cracking**, it's said to have good **plastic qualities**.
- 2) This could mean that a material is **malleable** (changes shape under pressure so can be moulded, e.g. by hammering) or **ductile** (can be drawn into wires).

BRITTLENESS:

- 1) **Brittle** materials can't withstand much **stretching**.
- 2) Brittle materials are more likely to **crack** or **break** than change their shape.
- 3) **Glass** and **acrylic** react like this under force.



TOUGHNESS:

- 1) **Tough** is the **opposite** of brittle.
- 2) If a material is tough, it's hard to **break** or **snap**.
- 3) **Armour** and bulletproof vests need to be tough.

DURABILITY:

- 1) If a product is **durable** it is able to withstand **repeated use**.
- 2) Durable products also withstand **wear and tear**, **weathering** and **corrosive attack**.



Wouldn't like to be that duck...

Not a bundle of laughs, sure, but darn important stuff. Learn it, then turn over the page and write down: 1) a definition of each of the six words 2) anything else interesting about that property.

PHYSICAL PROPERTIES

Physical properties are a major factor when selecting materials for specific users, purposes and conditions. Each physical property describes how a material behaves under specific conditions. Common physical properties are:

- **mechanical:** how a material will react to physical forces, such as how some materials become brittle when repeatedly bent or flexed
- **electrical:** a material's conductive properties including insulation and resistance, such as how a screwdriver's polymer handle (insulator) protects the user from electric shocks through the tip (conductor)
- **chemical:** how a material reacts to other chemicals, such as how metals oxidise when exposed (for example, ferrous metals rusting)
- **thermal:** how a material responds at different temperatures, such as freezing/melting and boiling points, such as the temperature ranges where thermoforming plastics can be shaped
- **magnetic:** most ferrous (containing iron) materials are attracted to magnets or can be magnetised, such as the use of magnetic catches for keeping doors closed
- **permeability:** the degree to which liquids or gases can pass through a material, such as woods that have been varnished having low permeability to liquids and water resistance
- **aesthetic:** the visual sensory properties of the surface/form of a material, such as colour and shine
- **tactile:** the touch or sensory properties of a material, such as texture.

WORKING PROPERTIES

Each **working property** has a technical definition describing how a material behaves when being worked or shaped (see unit 3.9). The most common working properties are:

- **strength:** the ability to withstand a constant external **force** without failing
- **hardness:** the ability to resist wear-and-tear, scratching and dents; usually related to surfaces
- **toughness:** the ability to withstand blows or momentary shocks (live forces)
- **durability:** the ability to resist wear, such as weathering, over time
- **elasticity:** the ability to bend and then to return to its original shape/size
- **plasticity:** the ability to permanently change shape or form, which applies to materials other than polymers
- **malleability:** the ability to permanently change shape, in all directions, while retaining integrity
- **ductility:** the ability to change shape by stretching along its length without snapping.

Choosing the Best Material to Use

Selecting the right materials isn't **easy**. But it's something you've got to be able to do, otherwise you can end up in a right **pickle**.

Different Factors affect your Selection of Material

You should be able to make a good choice as long as you understand a material's **properties**, and know what it's being **used** for.

Functional Requirements — What does the product have to do?

Ask yourself the following questions:

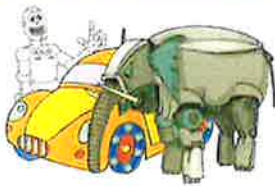
- 1) What **demands** will be made on the material? (Will it have to withstand heavy loads or chemicals? Will it have to conduct heat or electricity?)
- 2) Will it be for **outdoor** or **indoor** use? (If it's for outside, you'll need to consider whether your material will **corrode**.)
- 3) Does it need to fit in with an **environment**? (Your material might need to be a particular **style** or have a certain **look**.)

Most of this car seems to have rusted away, leaving just the boot and half the back wheel. Probably the result of a poor choice of material.



Economics — How much money have you got?

You'll need to consider the following points:



Nowadays most cars are assembled by robots.

- 1) The **size** of the product — materials like pewter are **expensive**, but may be a good choice for a small item of jewellery.
- 2) **Scale of production** — is your product a **one-off**, or will it be **batch** or **mass produced**? Stainless steel could be a possible material for a one-off product. But if you're mass-producing something that would be equally as good made from some kind of plastic, that might be cheaper.

Availability of Supply — What can you get hold of?

Can you get hold of the material you want? And if you can, can you get it in a suitable form? Most materials are only available in **standard forms** and **sizes**, and it can be **very expensive** to get a material in any other form. This will have a direct effect on the **cost** and the **method of manufacture**.

For example, materials might be available as granules, strips, bars, tubes, rough sawn, planed...

Manufacturing Method — How will the product be made?

- 1) Some materials are easier to **join** than others (which will affect the production method used).
- 2) Also, the **material** must be suitable for the intended **production method** (and vice versa). For example, you can make something out of certain plastics using **injection moulding**, but it's no good planning to use this technique for wood.

I reckon you should just make everything out of cheese...

You need to know about different materials so you can choose the right one for the job in hand. Don't make a fireguard or teapot from chocolate, for example — they won't be much use. It sounds obvious, but it's a mistake that amateurs make all too often. Be smart — don't be one of them.

3.7 Preparation of materials

LEARNING OBJECTIVES

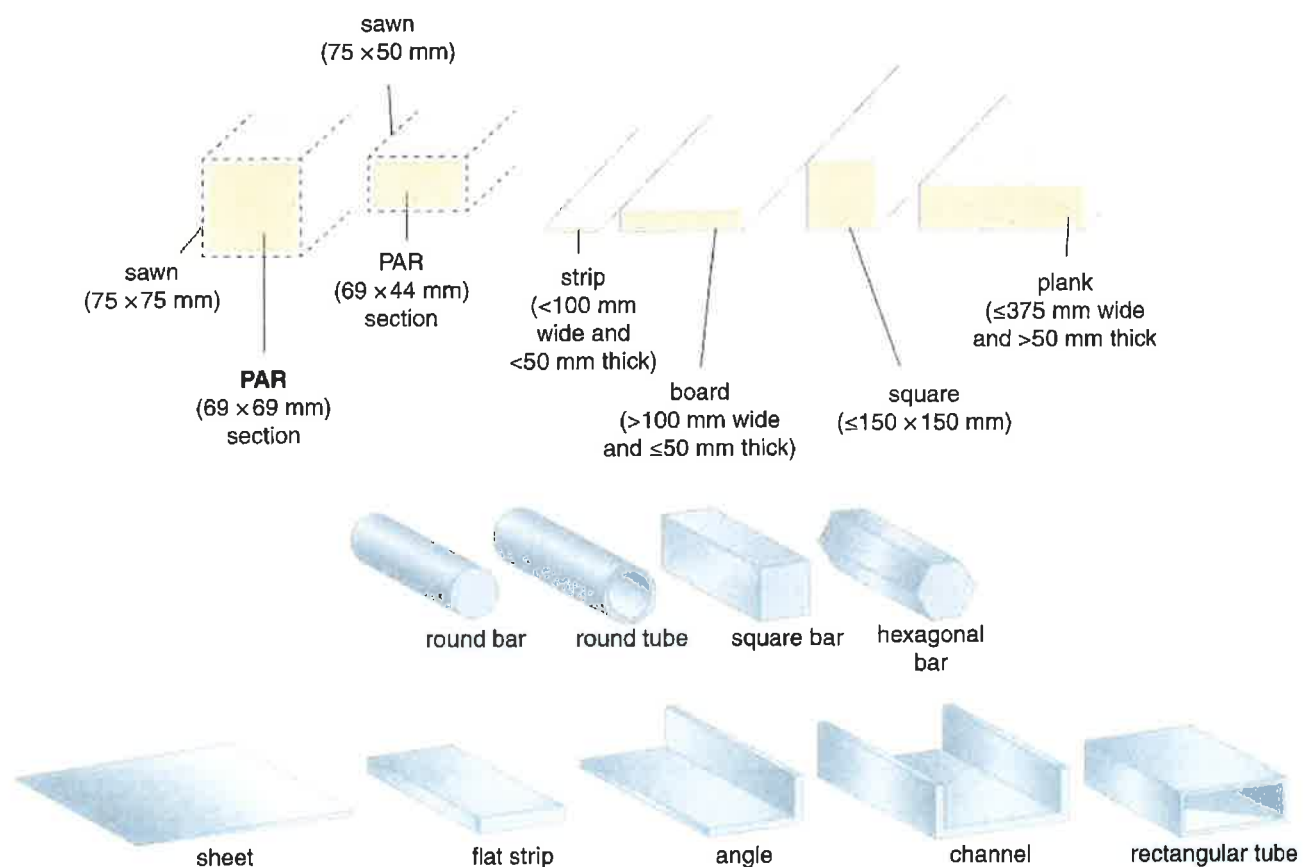
By the end of this unit you should:

- ✓ be able to demonstrate knowledge of available market forms, types and sizes
- ✓ understand different methods of cutting
- ✓ understand the use of datum surfaces, lines and edges and know how to produce them
- ✓ be able to explain the preparation needed for machine processes and safe methods of securing materials to work surfaces.

WHAT IS MARKET FORM?

Market form describes manufactured standard shapes and sizes of material, wood or metal available to buy commercially.

When planning a project, knowledge of market forms, shapes and sizes of material has several advantages. Effective use of this knowledge helps to ensure that project costs are kept to a minimum, particularly when working in metal. By using pre-cut material, time is saved and subsequently labour costs are reduced. Using market forms is also better for the environment because it means there is less waste material. This illustration shows industry standard forms, shapes and sizes for metal and wood.



△ Examples of standard sectional profiles available in different materials.